

CS & IT ENGINEERING

Digital Logic

Sequential Circuit

Lecture No. 7



By- Aditya sir

Recap of Previous Lecture



Topic



Design of FF
Register

SISO

SIPO

PISO

PIPO

Topics to be Covered



Topic

Sequential Circuit

- Questions on Register
- Toggle Mode of a FF
- Counter





About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



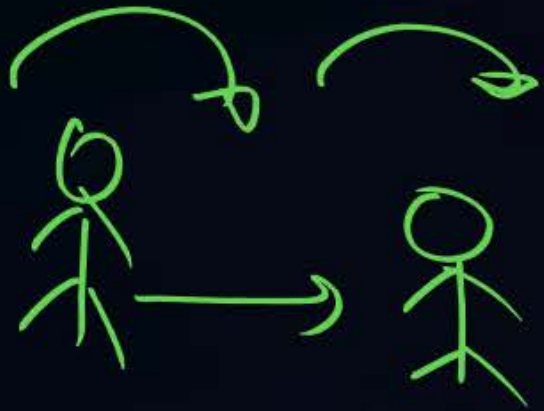


Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Sequential Circuit

Summary



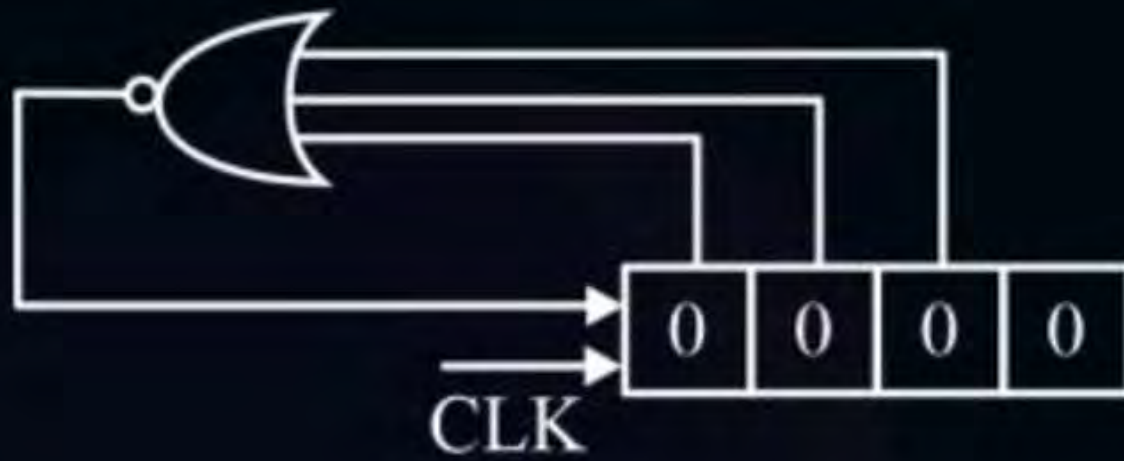
	↓ Retrieval			
	Store	Relative	Total	
SISO	n	$n-1$	$(2n-1)$	→ Slowest
SIPO	n	0	<u>n</u>	
PISO	1	$n-1$	n	→ $1+n-1 = \textcircled{n}$
PIPO	1	0	1	→ Fastest

Serial → Temporal code
Parallel → Spatial code

[MCQ]



#Q. Write the state of the register given below?
(Assume initially all flips flops are reset)

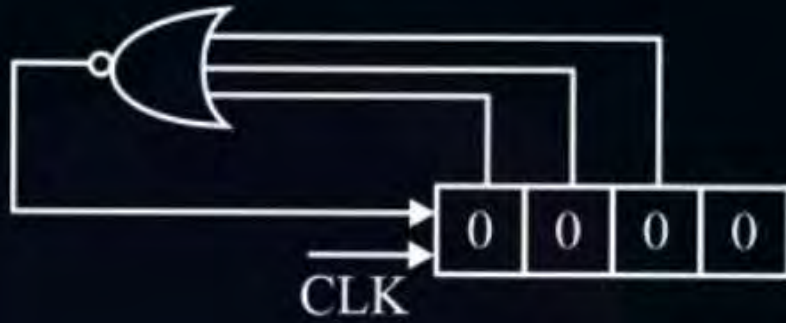


Clock	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0
6	0	1	0	0
7	0	0	1	0
8	0	0	0	1
c				

[MCQ]



#Q. Write the state of the register given below?
(Assume initially all flips flops are reset)



Clock	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0
6	0	1	0	0
7	0	0	1	0
8	0	0	0	1
9	1	0	0	0

10 0 1 0 0
11 0 0 1 0

o/p after 50th clock? $\longrightarrow$ 0100

1 $\longrightarrow$ 5 $\longrightarrow$ 9	57:4 = 1
$\longrightarrow$ 2 $\longrightarrow$ 6 $\longrightarrow$ 10	67:4 = 2
3 $\longrightarrow$ 7 $\longrightarrow$ 11	77:4 = 3
4 $\longrightarrow$ 8 $\longrightarrow$ 12	87:4 = 0

$$507:4 = (2)$$

o/p at 60th clock? $\longrightarrow$ 0001

$$607:4 = 0$$

after 1257th clock?

$\downarrow$

(1209) + 57

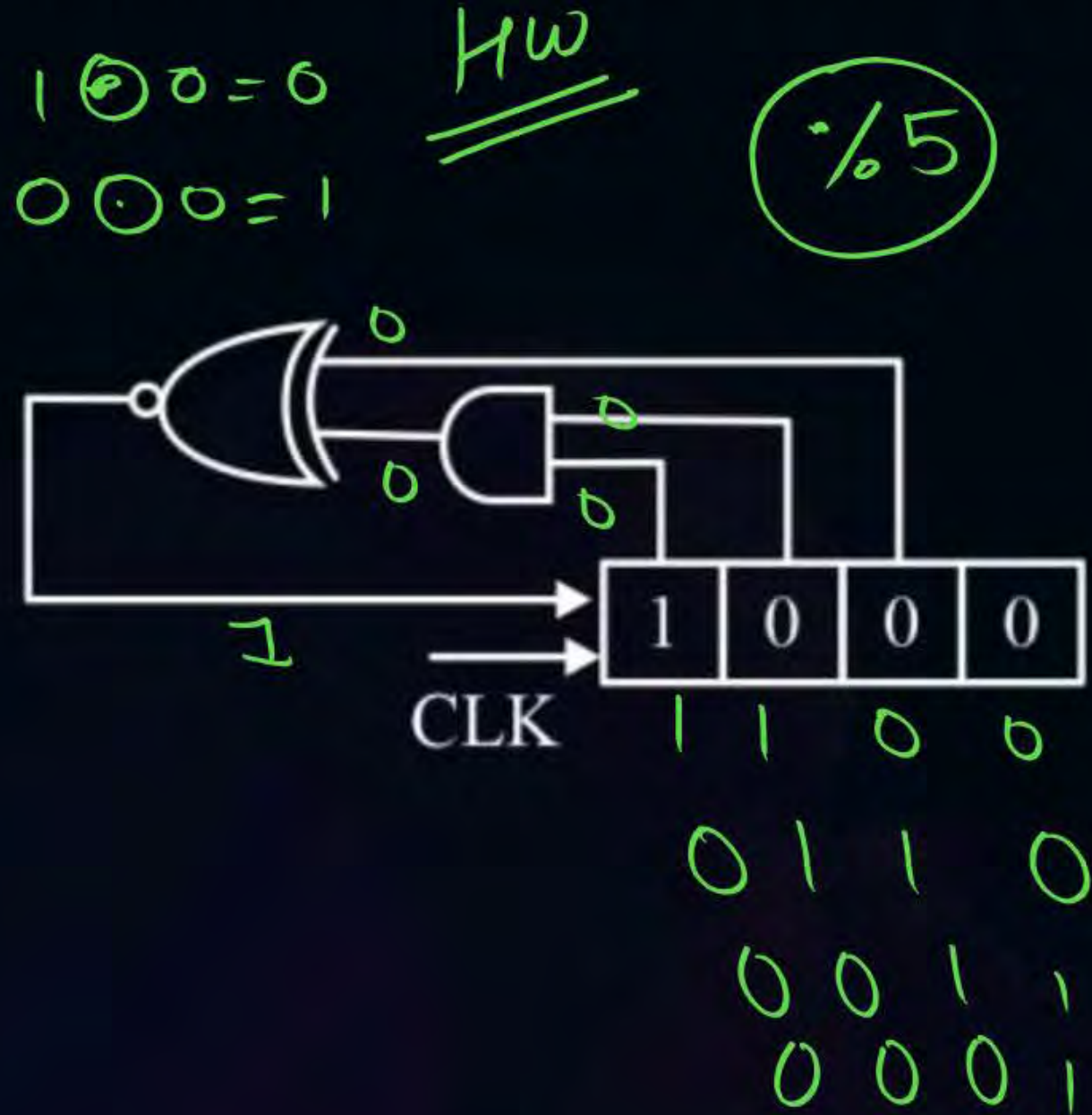
$\longrightarrow$ 1000 ✓

$$577:4 \longrightarrow 1$$

[NAT]



#Q. After 97th clock the output will be?



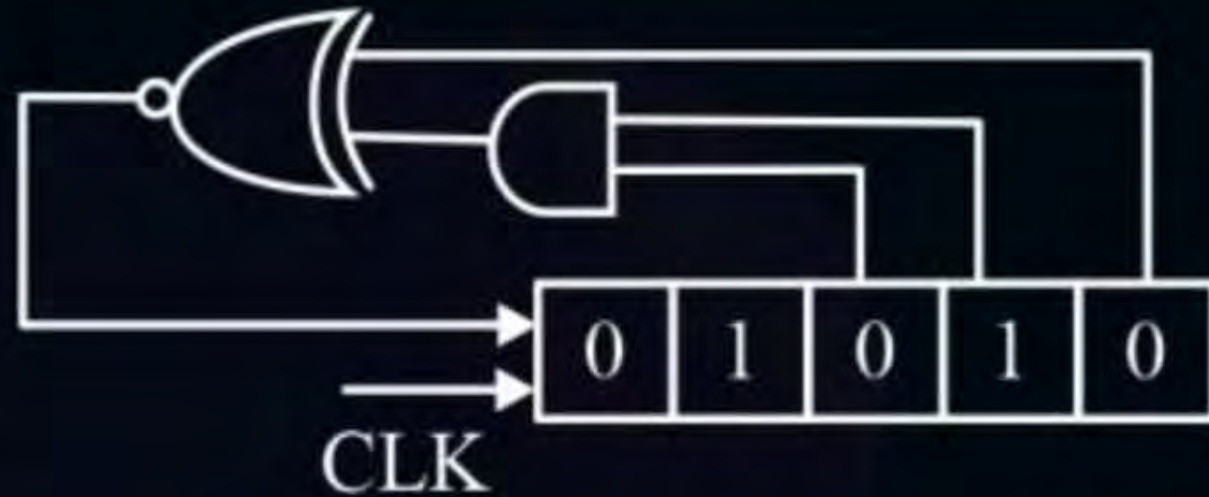
Clock	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1
4	0	0	0	1
5	1	0	0	0
6	1	1	0	0
7				
8				
9				
10				
11				
12				
13				
14				

[NAT]



#Q. After 137th clock the output will be?

1402



Clock	Q ₃	Q ₂	Q ₁	Q ₀
0				
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				

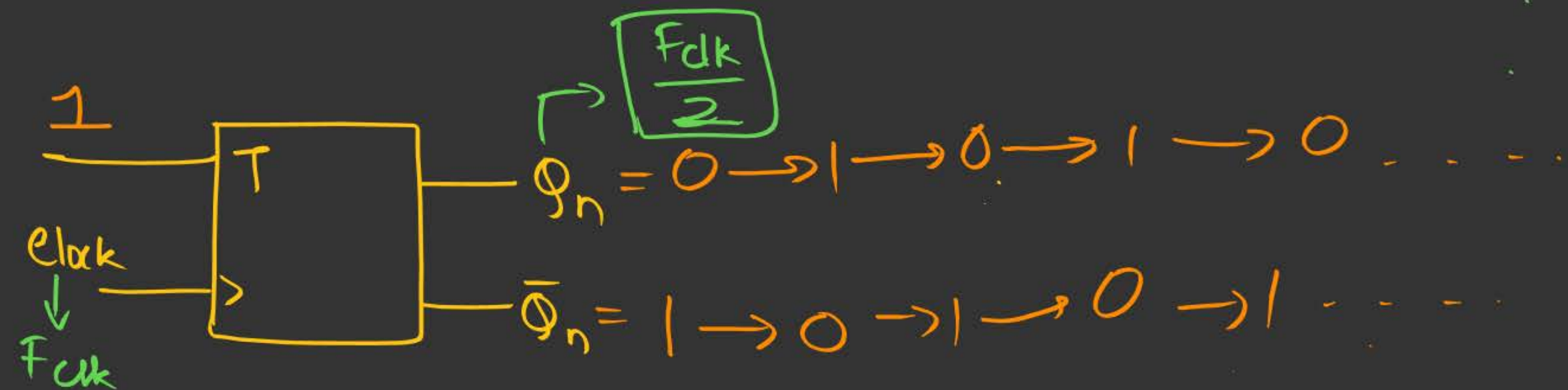
Toggle Mode of a FF

$T = 1 \rightarrow \underline{\text{Toggle}}$

$JK \rightarrow \begin{matrix} J=1 \\ K=1 \end{matrix} \left\{ \begin{matrix} \text{Toggle} \end{matrix} \right.$

Toggle: $Q_{n+1} = \overline{Q_n}$

② Toggle mode of T FF



validate:

$$Q_{n+1} = T \oplus Q_n$$

for T=1

$$= 1 \oplus Q_n$$

$$= \overline{Q_n}$$

$$Q_{n+1} = \overline{Q_n}$$

Bonus:

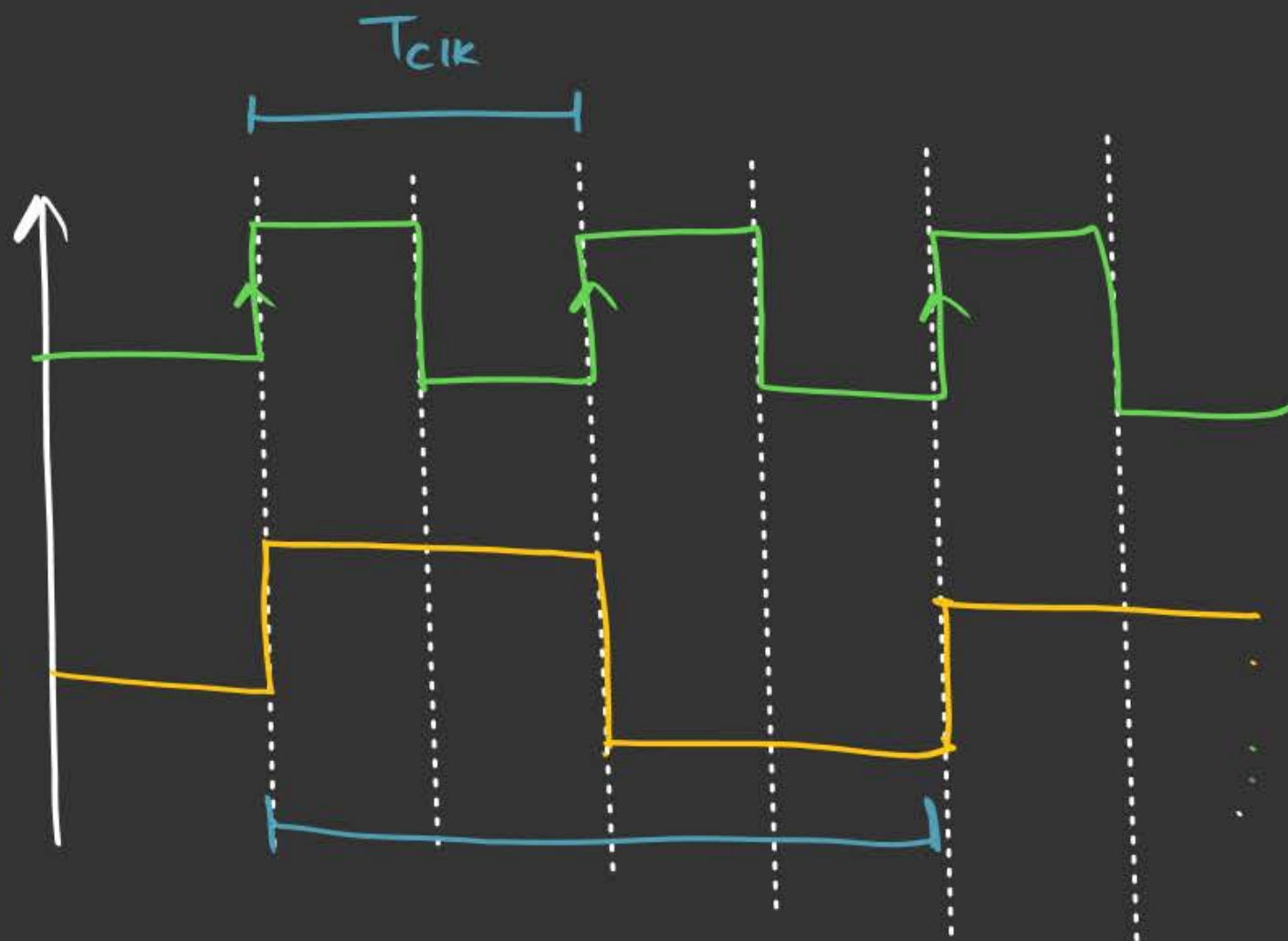
T_{FF}

$Q_n = 0$

Clock

Q_{n+1}

$T_{Q_{n+1}} = 2 * T_{clk}$



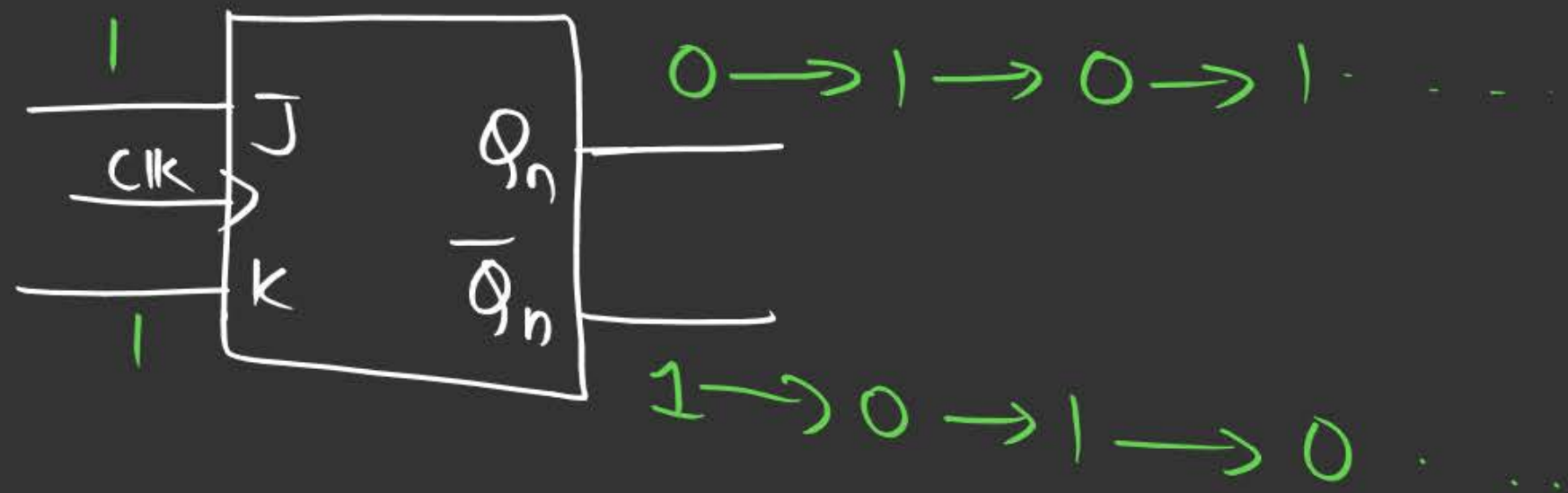
$$f_{Q_{n+1}} = \frac{f_{clk}}{2}$$

frequency

$$f_{clk} = \frac{1}{T_{clk}}$$

$$f_{Q_{n+1}} = \frac{1}{T_{Q_{n+1}}} = \frac{1}{2T_{clk}} = \boxed{\frac{f_{clk}}{2}}$$

② Toggle mode of a JK FF



validate:

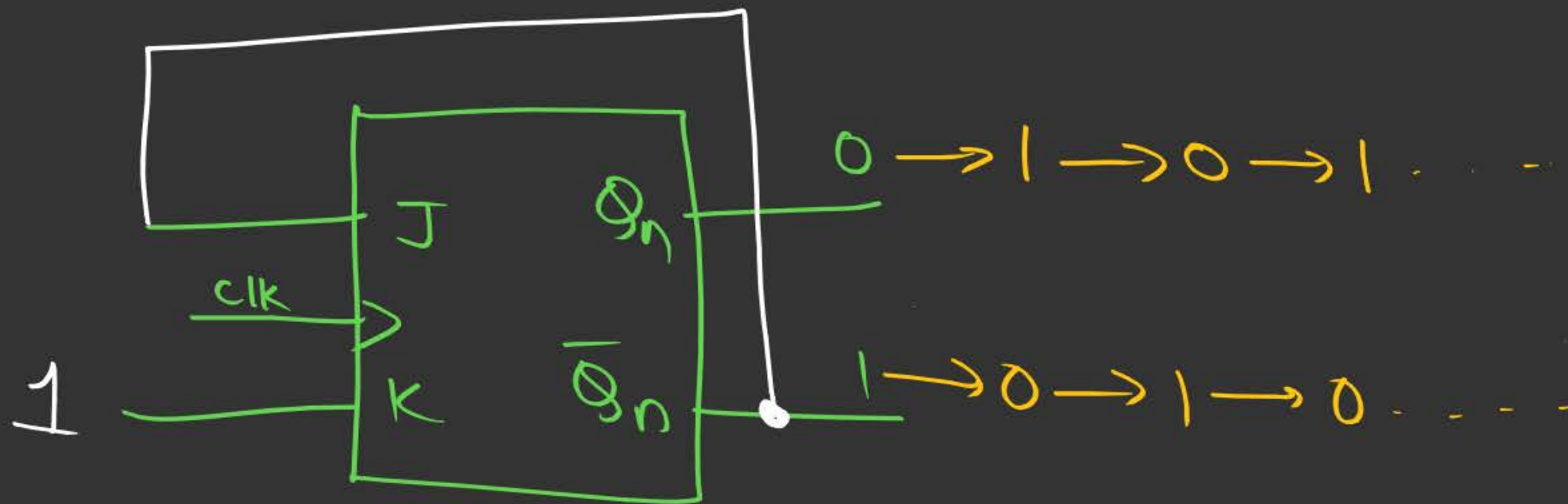
Put $J=1$
 $K=1$

$$Q_{n+1} = J \overline{Q}_n + \overline{K} Q_n$$

$$Q_{n+1} = 1 \cdot \overline{Q}_n + \overline{1} \cdot Q_n$$

$$Q_{n+1} = \overline{Q}_n$$

③ Toggle mode of a JK FF



$J=0$
 $K=1$ } → Reset
 $Q_n=0$
 $\bar{Q}_n=1$

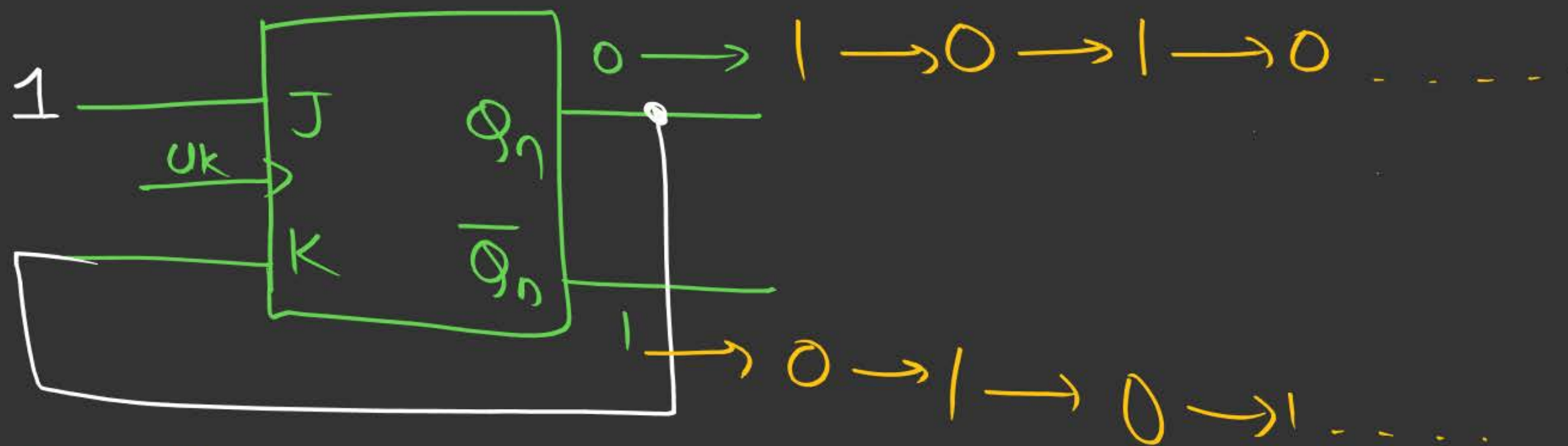
validate prov FF:

$$\left. \begin{array}{l} J = \overline{Q_n} \\ K = 1 \end{array} \right\}$$

$$\begin{aligned} Q_{n+1} &= \overline{J} Q_n + \overline{K} Q_n \\ &= \overline{Q_n} \cdot \overline{Q_n} + \overline{1} \cdot Q_n \\ &= \overline{Q_n} + 0 \end{aligned}$$

$$\boxed{Q_{n+1} = \overline{Q_n}}$$

④ Toggle mode of JK FF



$J=1$
 $K=Q_n$

$\Rightarrow$ Set
 $Q_n=1$
 $Q_{\overline{n}}=0$

validate

$$\left. \begin{array}{l} J=1 \\ K=Q_n \end{array} \right\} \rightarrow$$

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

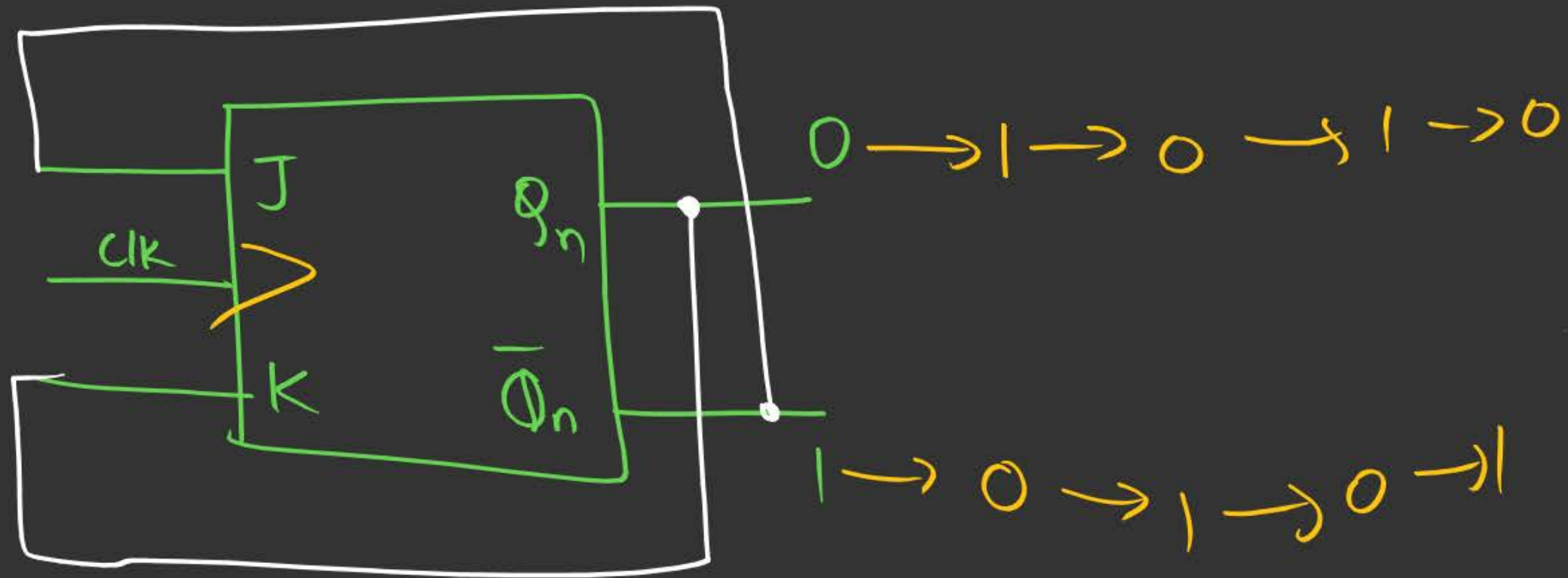
$$= 1 \cdot \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$= \bar{Q}_n + 0$$

$$\boxed{Q_{n+1} = \bar{Q}_n}$$



⑤ Toggle mode of a JK FF.



J K

1 0 → Set

0 1 → Reset

validate:

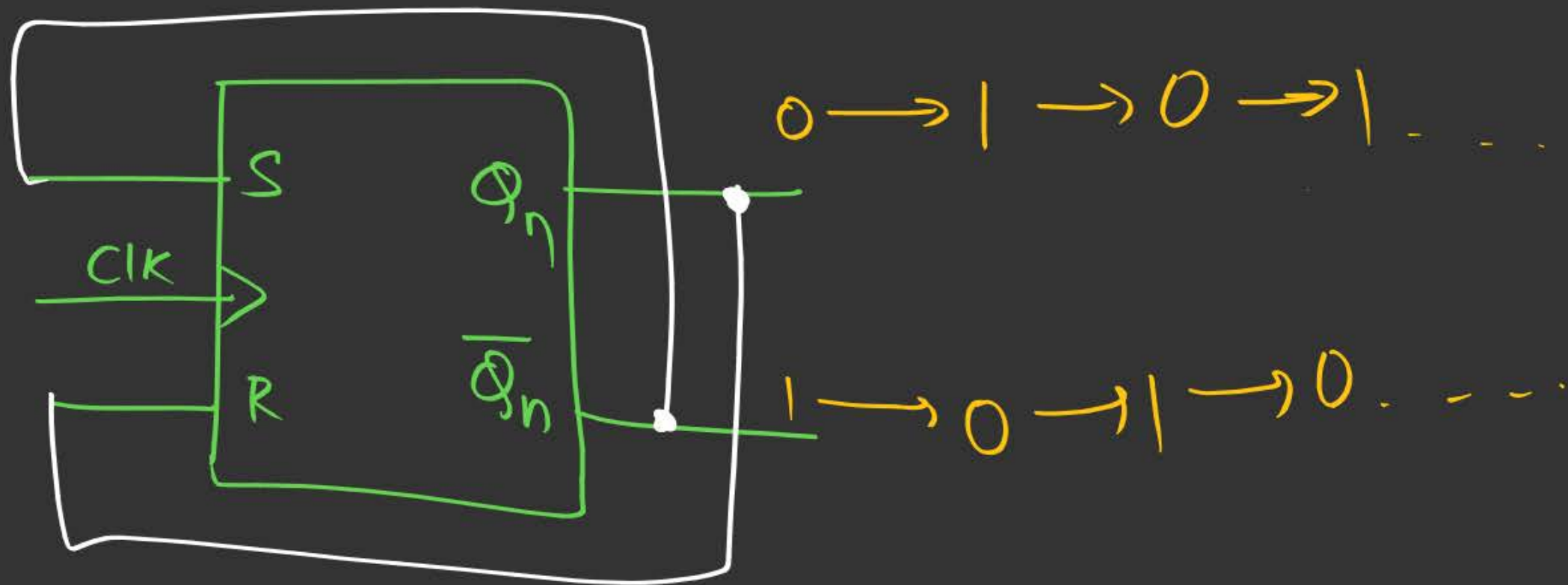
$$\left. \begin{array}{l} J = \bar{Q}_n \\ K = Q_n \end{array} \right\}$$

$$Q_{n+1} = J \bar{Q}_n + \bar{K} Q_n$$

$$= \bar{Q}_n \cdot \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$\boxed{Q_{n+1} = \bar{Q}_n}$$

⑥ Toggle mode of SR FF.



S R
1 0 $\rightarrow$ Set
0 1 $\rightarrow$ Reset

validate

$$S = \bar{Q}_n$$

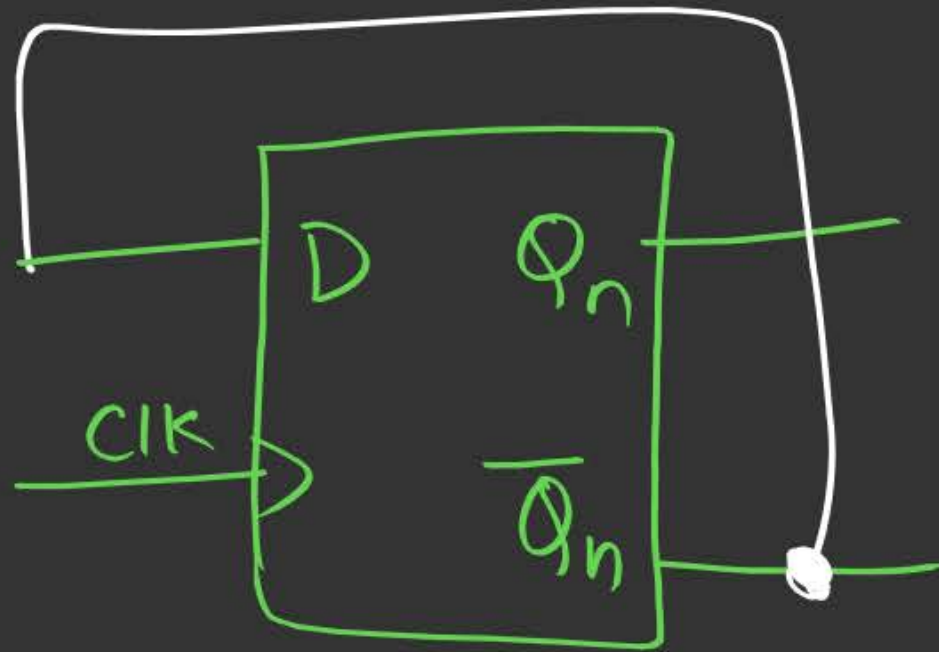
$$R = Q_n$$

$$Q_{n+1} = S + \bar{R} \cdot Q_n$$

$$= \bar{Q}_n + \bar{Q}_n \cdot Q_n$$

$$\boxed{Q_{n+1} = \bar{Q}_n} \checkmark$$

⑦ Toggle mode of a D FF



0 → 1 → 0 → 1

1 → 0 → 1 → 0

Validate

$$D = \overline{Q}_n$$

$$Q_{n+1} = D$$

$$Q_{n+1} = \overline{Q}_n$$



Summary:-

Total there are only 7 Toggle modes of FF

$$T \rightarrow ① \Rightarrow \boxed{T=1}$$

$$JK \rightarrow ④ \rightarrow \boxed{J=1, k=1}, \boxed{J=\bar{Q}_n, k=1}, \boxed{J=1, k=Q_n}, \boxed{J=\bar{Q}_n, k=Q_n}$$

$$SR \rightarrow ① \rightarrow \boxed{S=\bar{Q}_n, R=Q_n}$$

$$D \rightarrow ① \rightarrow \boxed{D=\bar{Q}_n}$$

Counters



Topic : Counter

1. Counters are used to count Number of clock. ★
2. Counter are used as Frequency Divider Circuit.
3. Counter are also used in ADC.
4. Counters are also known as pulse stretcher circuit.
5. Counters also used in RADAR for detection of Range.

↓
Radio Detection & Ranging



Topic : Types of Counter

1. Asynchronous Counter.

2. Synchronous Counter.



Topic : Counter

1. Asynchronous Counter:

Only one flip flop having external clock and output of that flip flop will become clock for the next flip flop.

So when clock applied only one flip flop work at that time.



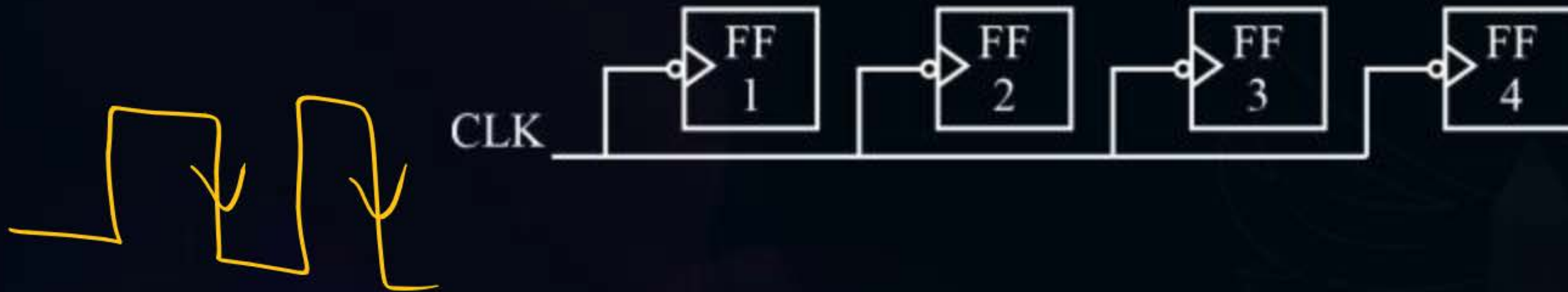
(default)



Topic : Counter

2. Synchronous Counter:

All Flip Flops are connected with the same clock. Hence when clock is applied all the flip flops work simultaneously.





Topic : Counter

Imp Differences

Synchronous Counter	Asynchronous Counter
(1) All the FF's are connected with the <u>same</u> clock.	(1) <u>Only one</u> FF having external clock and o/p of that FF will be clock for the next FF.
(2) Faster → → → →	(2) Slower ↑ ↑ ↑ ↑
(3) <u>All type</u> of counting are possible.	(3) Only <u>increasing</u> and <u>decreasing</u> counting are possible.
(4) <u>No</u> , Transition Error.	(4) <u>Yes</u> , Transition Error.
(5) Ex: <u>Ring counter</u> , <u>Johnson counter</u>	(5) Ex: <u>Ripple counter</u>

Types of Counting:

- ① Increasing: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \dots$
- ② Decreasing: $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$
- ③ Random: $5 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 4 \dots$



2 mins Summary



- Register
- Toggle modes
- Intro to Counters



THANK - YOU